# Agentic Procurement & Working Capital Management for Supply Chain Finance

Where deterministic finance meets autonomous intelligence

# COIMBATORE INSTITUTE OF TECHNOLOGY

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# The Problem

Modern supply chains suffer from a critical **trust-liquidity paradox:**

Businesses lock up **$2.3 trillion** globally in working capital due to **payment timing mismatches** between buyers and sellers. Traditional solutions fail because:

**Identity Trust Gap:** Buyers cannot **verify seller credentials cryptographically** → manual KYC delays, fraud risk

**Payment Timing Mismatch:** Buyers want to delay payment **(preserve cash)**, sellers need early payment **(cash flow)** → no mechanism to optimize for both

**Opaque Cash Flows:** Companies cannot predict **liquidity needs** accurately → over-reserve cash, miss yield opportunities

**Fragmented Systems:** Invoice issuance, credit scoring, payment negotiation, treasury management operate in silos → **suboptimal decisions**

# The Core Problem is... ?

The problem is not **lack of money.**

The problem is lack of **verifiable trust + intelligent timing + integrated decision-making.**

**Until supply chains can:**

- Cryptographically verify who is acting
- Deterministically know what must be paid
- Intelligently predict when risk and liquidity change

...working capital will **remain trapped.**

# Coventional Vs Novel Approach

## Conventional Systems

- Manual decision-making (finance teams)
- Rule-based AR/AP management
- Static cash flow tracking (no prediction)
- Basic credit scoring (historical data)
- Idle cash = no yield optimization

## Our Novel System

- Agentic AI → Autonomous Buyer, Seller, Treasury agents
- ACTUS Engine → 100% deterministic cash flow modeling
- AI Forecasting → Predict risk, behavior & market changes
- Dynamic Discount Optimization → Smart payment decisions
- DeFi Yield Layer → Earn on idle treasury funds
- Compliance-Aware AI → Basel-aligned + explainable

# Our Solution

## Trust-Driven Intelligent Supply Chain Finance

**Cryptographic Identity :**
Buyers, sellers, and AI agents are verifiably authenticated

**Smart Invoices :**
Invoices are converted into deterministic cash flow contracts

**AI Intelligence:**
Predicts payment delays, credit risk, and liquidity needs

**Agentic Negotiation :**
Buyer & seller agents optimize payment timing dynamically

**Treasury Optimization:**
Excess cash is deployed while maintaining liquidity buffers

# Data Acquisition

## 1. ACTUS Definitions & Concepts

- Standardized contract types (PAM, ANN, FXOUT)
- Financial attributes: principal, rate, maturity, event schedules
- Deterministic state transition & cash flow logic
- Provides mathematical foundation for accurate cash flow generation.

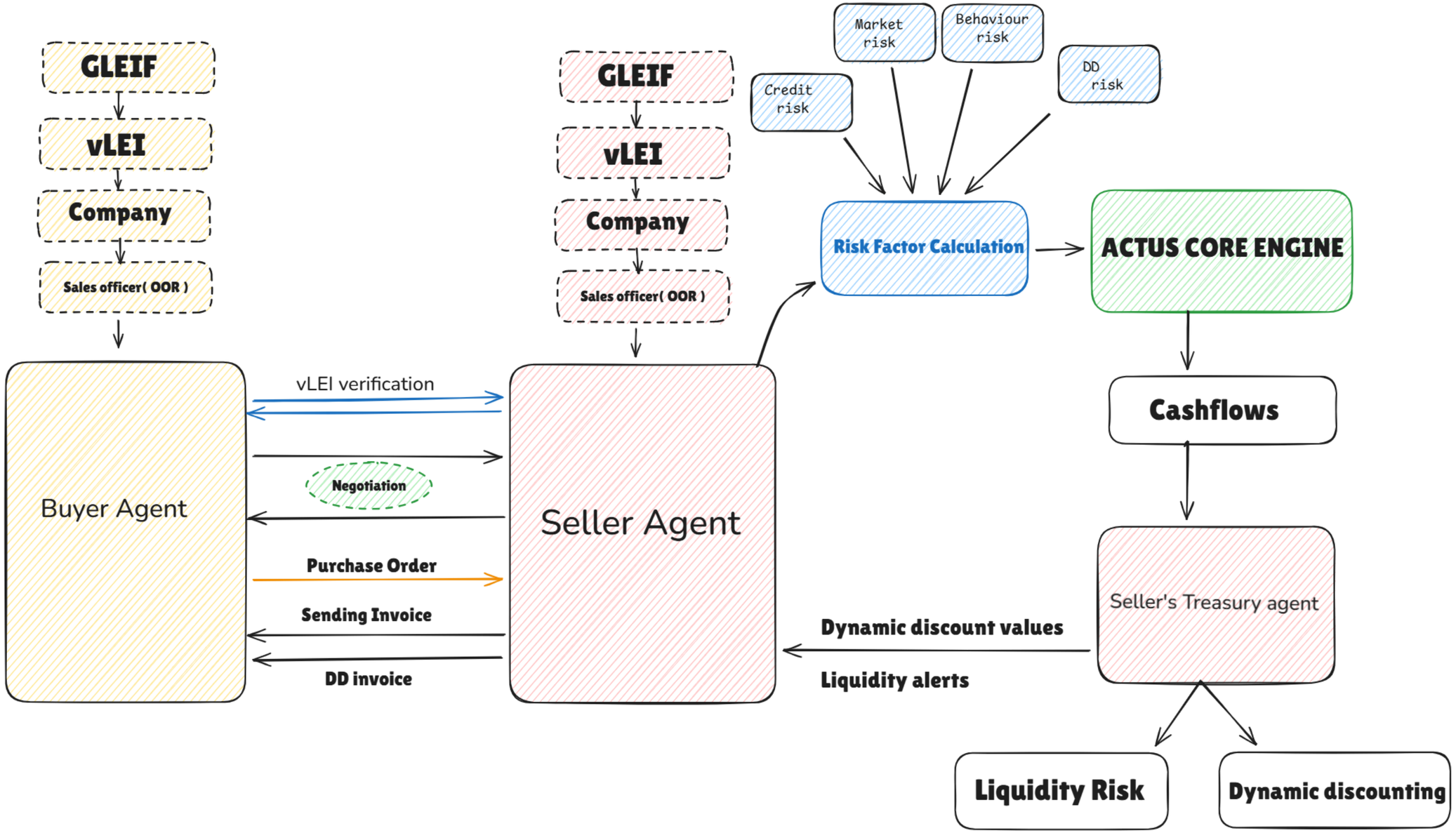
## 2. ACTUS Example Contracts

- Sample invoices, loans, installment contracts, FX trades
- Multiple maturity dates, interest structures, and payment patterns
- Enables multi-scenario simulations for AR/AP portfolio modeling.

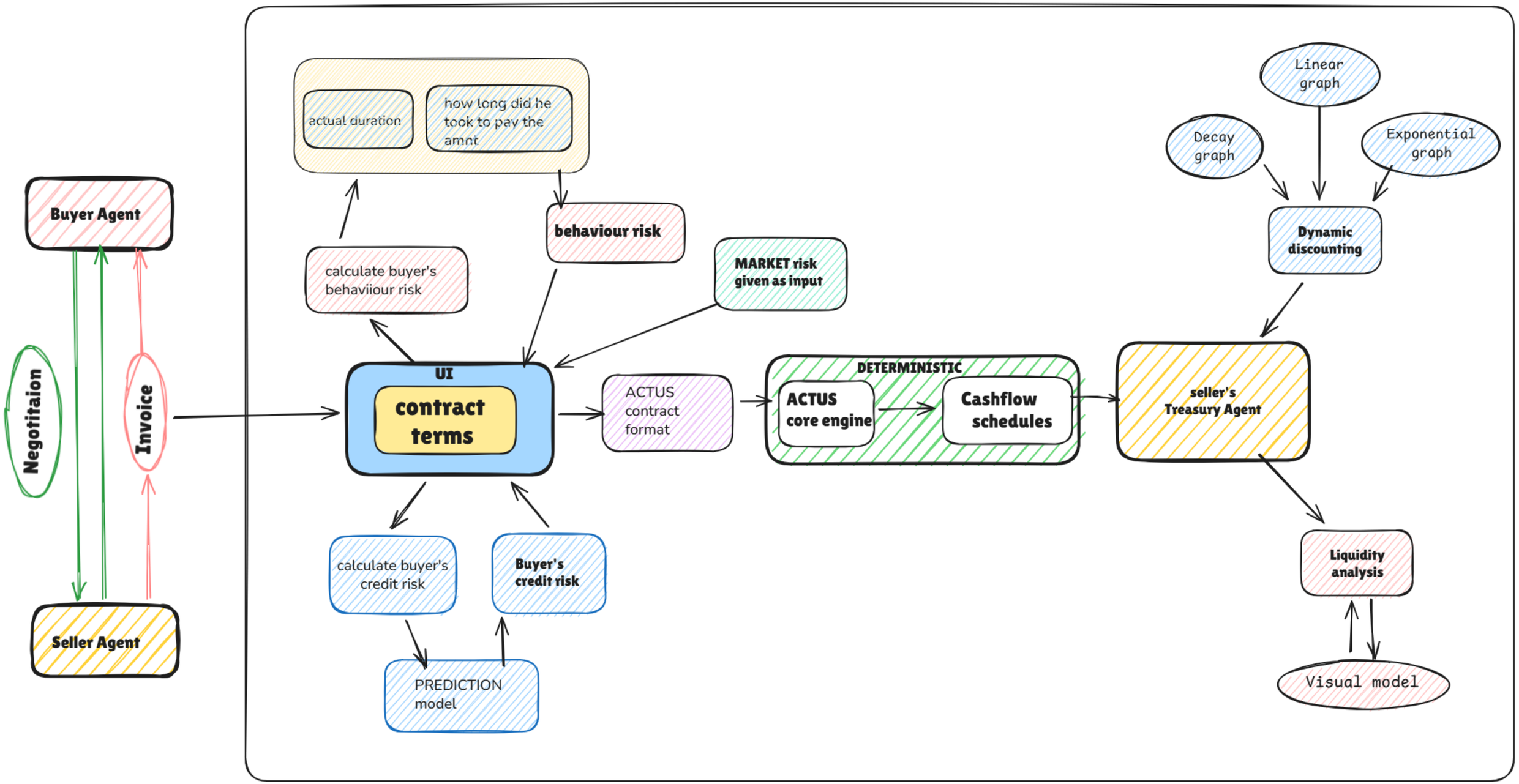
## 3. Executable contract Collection

- API endpoints for contract creation & cash flow execution
- Automated testing of contract outputs
- Ensures 100% deterministic execution before AI-based risk forecasting.

# Architecture



# Flow Diagram



# THANK YOU